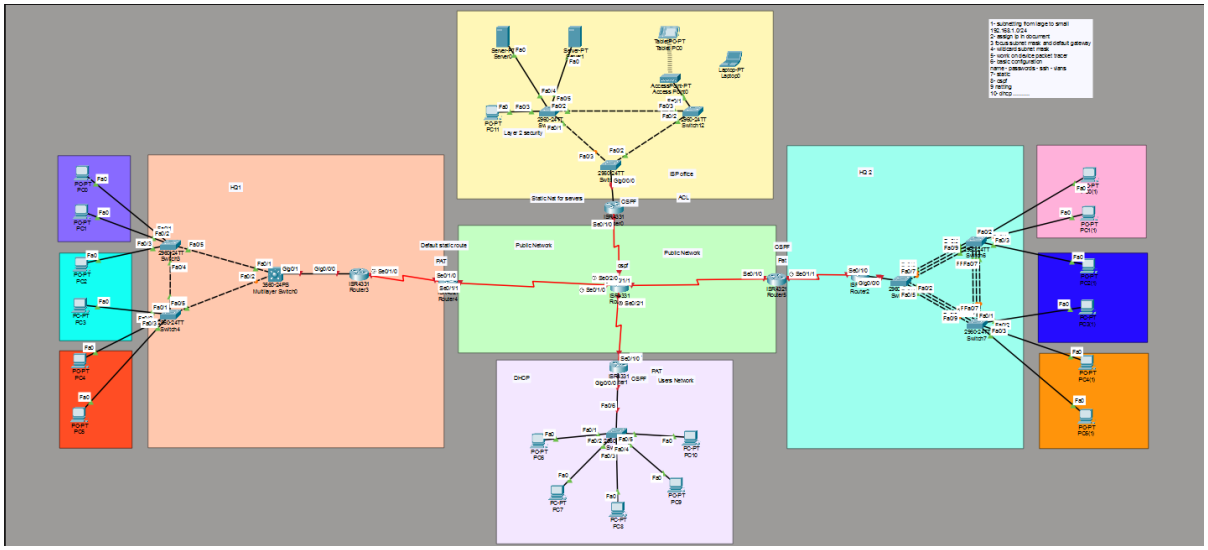
Supply Chain Network



Prepared by: Haneen Nasr Hosny

Under the Supervision of: Saher Waleed

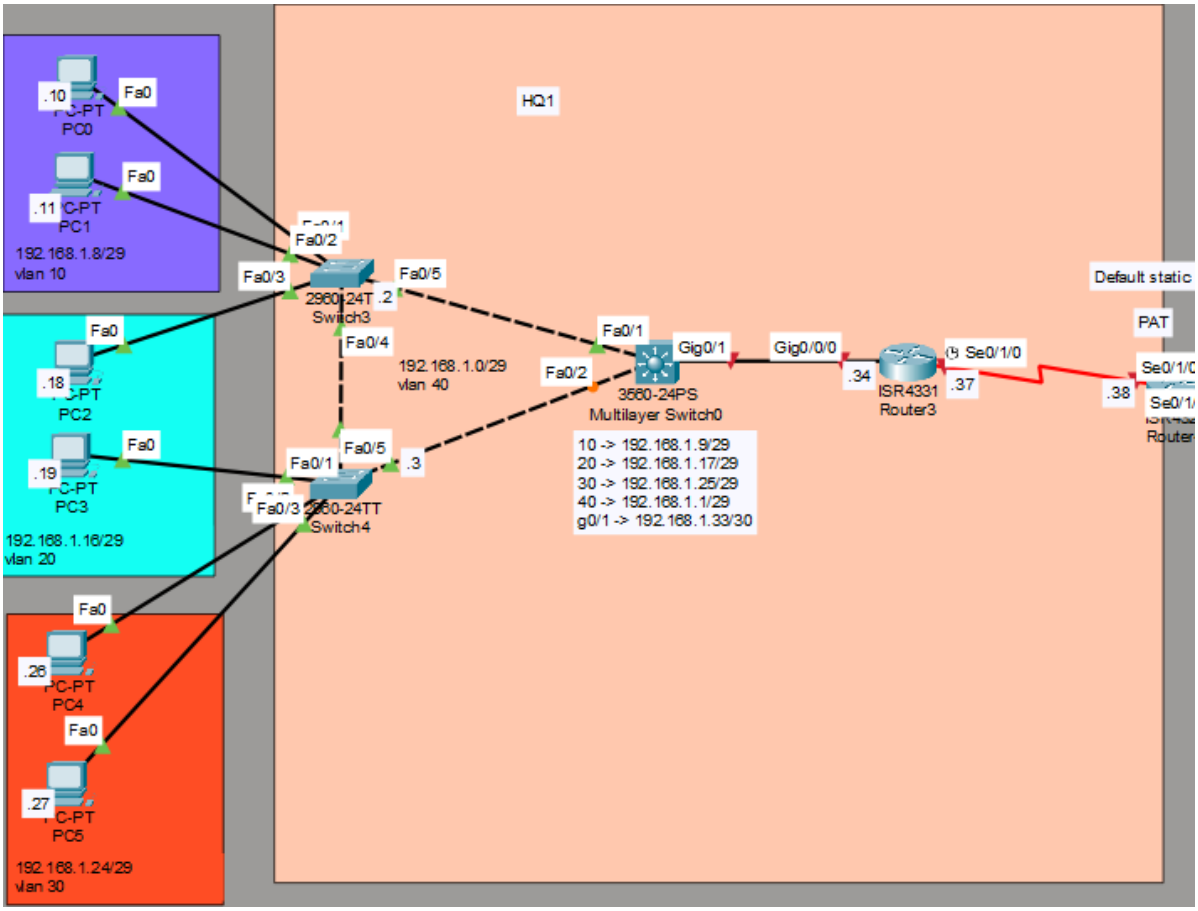
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Project Goals

- Use VLANs to separate departments and keep traffic organized.
- Use EtherChannel (LACP) to combine links so the network stays up if one link fails.
- Use multilayer switches to route between VLANs and give each subnet a gateway.
- Set up OSPF with a default route so the network updates itself automatically.
- Give internal devices internet access using Dynamic PAT (NAT Overload).
- Make internal servers reachable from outside using Static NAT.
- Add security and remote access to protect and manage the devices.

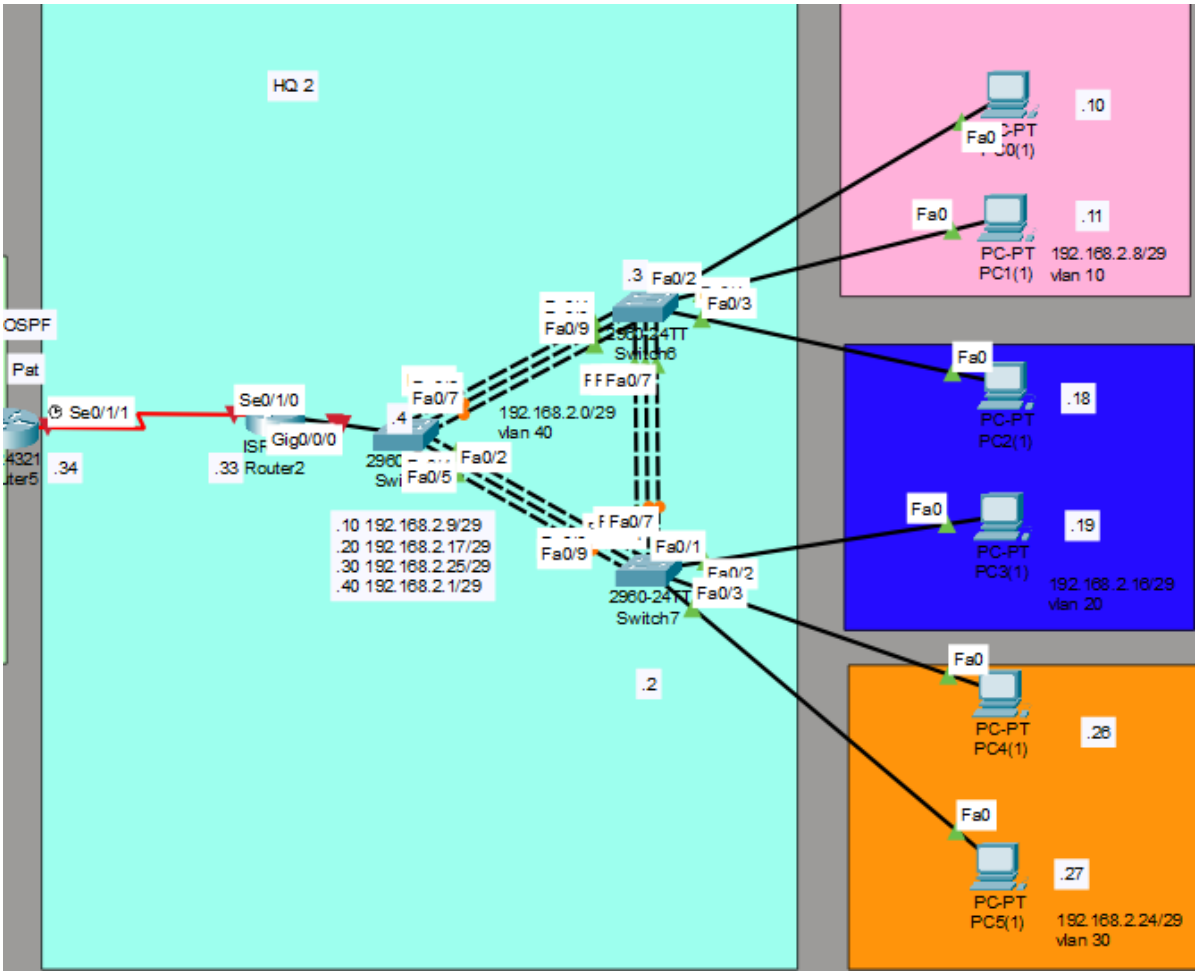
Subnetting & Device Tables



HQ1 – VLANs 10/20/30 and Multilayer Switch0

Device	Interface	IP Address	Subnet Mask	Wildcard Mask
PC0	Fa0	192.168.1.10	255.255.255.248	0.0.0.7
PC1	Fa0	192.168.1.11	255.255.255.248	0.0.0.7
PC2	Fa0	192.168.1.18	255.255.255.248	0.0.0.7
PC3	Fa0	192.168.1.19	255.255.255.248	0.0.0.7
PC4	Fa0	192.168.1.26	255.255.255.248	0.0.0.7
PC5	Fa0	192.168.1.27	255.255.255.248	0.0.0.7
Multilayer Switch0	VLAN 10 (SVI)	192.168.1.9	255.255.255.248	0.0.0.7
Multilayer Switch0	VLAN 20 (SVI)	192.168.1.17	255.255.255.248	0.0.0.7
Multilayer Switch0	VLAN 30 (SVI)	192.168.1.25	255.255.255.248	0.0.0.7
Multilayer Switch0	VLAN 40 (SVI)	192.168.1.1	255.255.255.248	0.0.0.7
Switch3	VLAN 40	192.168.1.2	255.255.255.248	0.0.0.7
Switch4	VLAN 40	192.168.1.3	255.255.255.248	0.0.0.7
Multilayer Switch0	Gig0/1	192.168.1.33	255.255.255.252	0.0.0.3
Router3 (ISR4331)	Gig0/0/0	192.168.1.34	255.255.255.252	0.0.0.3
Router3 (ISR4331)	Se0/1/0	192.168.1.37	255.255.255.252	0.0.0.3

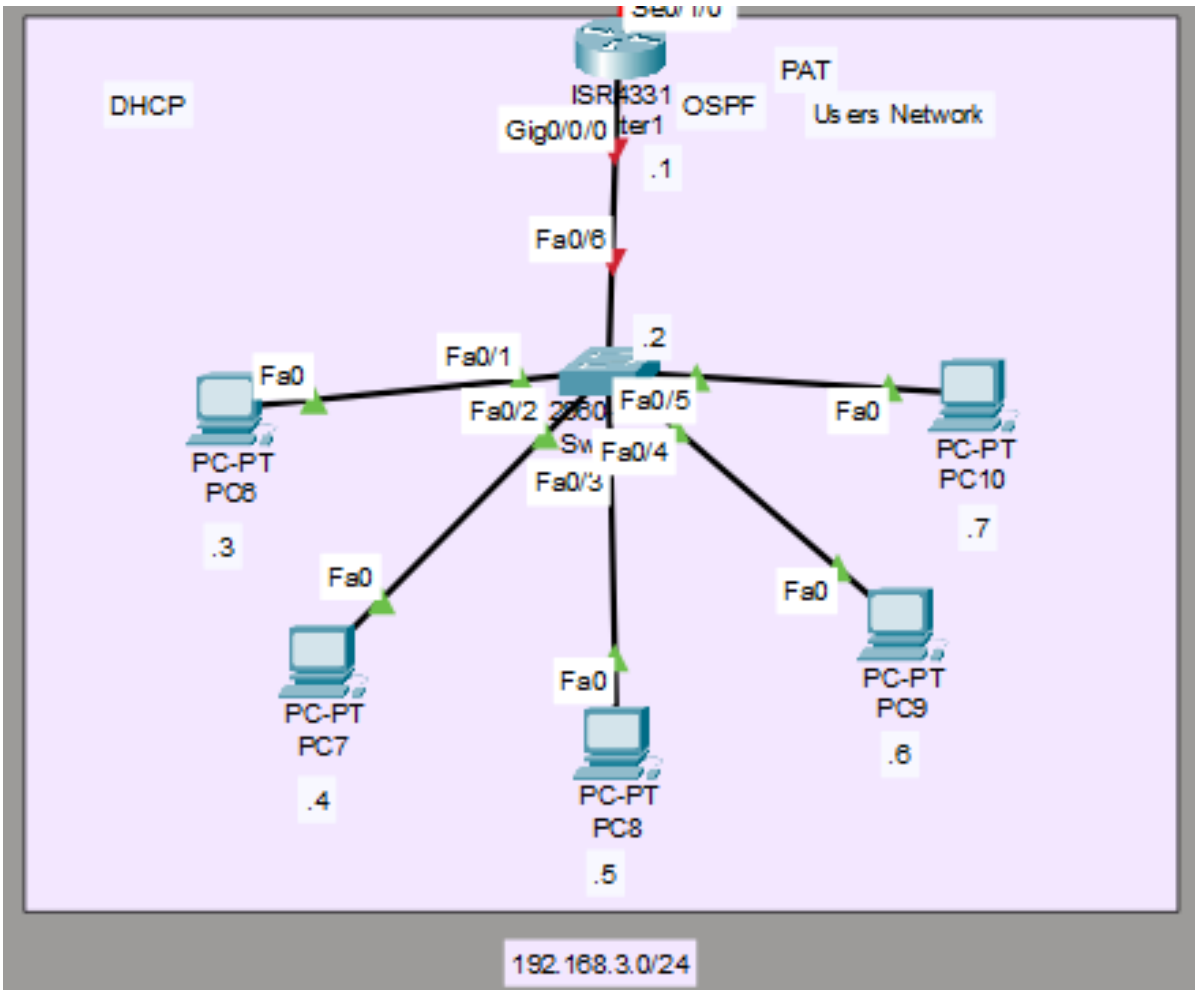
PC0-PC5 are split into VLANs 10, 20, and 30 through Switch3 and Switch4. Both trunk up to Multilayer Switch0, which routes between the VLANs and connects to Router3.



HQ2 – VLANs 10/20/30 (Router2 as Gateway)

Device	Interface	IP Address	Subnet Mask	Wildcard Mask
PC0(1)	Fa0	192.168.2.10	255.255.255.248	0.0.0.7
PC1(1)	Fa0	192.168.2.11	255.255.255.248	0.0.0.7
PC2(1)	Fa0	192.168.2.18	255.255.255.248	0.0.0.7
PC3(1)	Fa0	192.168.2.19	255.255.255.248	0.0.0.7
PC4(1)	Fa0	192.168.2.26	255.255.255.248	0.0.0.7
PC5(1)	Fa0	192.168.2.27	255.255.255.248	0.0.0.7
Router2 (ISR)	Gig0/0/0.10	192.168.2.9	255.255.255.248	0.0.0.7
Router2 (ISR)	Gig0/0/0.20	192.168.2.17	255.255.255.248	0.0.0.7
Router2 (ISR)	Gig0/0/0.30	192.168.2.25	255.255.255.248	0.0.0.7
Router2 (ISR)	Gig0/0/0.40	192.168.2.1	255.255.255.248	0.0.0.7
Switch7	VLAN 40	192.168.2.2	255.255.255.248	0.0.0.7
Switch6	VLAN 40	192.168.2.3	255.255.255.248	0.0.0.7
Switch0	VLAN 40	192.168.2.4	255.255.255.248	0.0.0.7
Router2 (ISR)	Se0/1/0	192.168.2.33	255.255.255.252	0.0.0.3

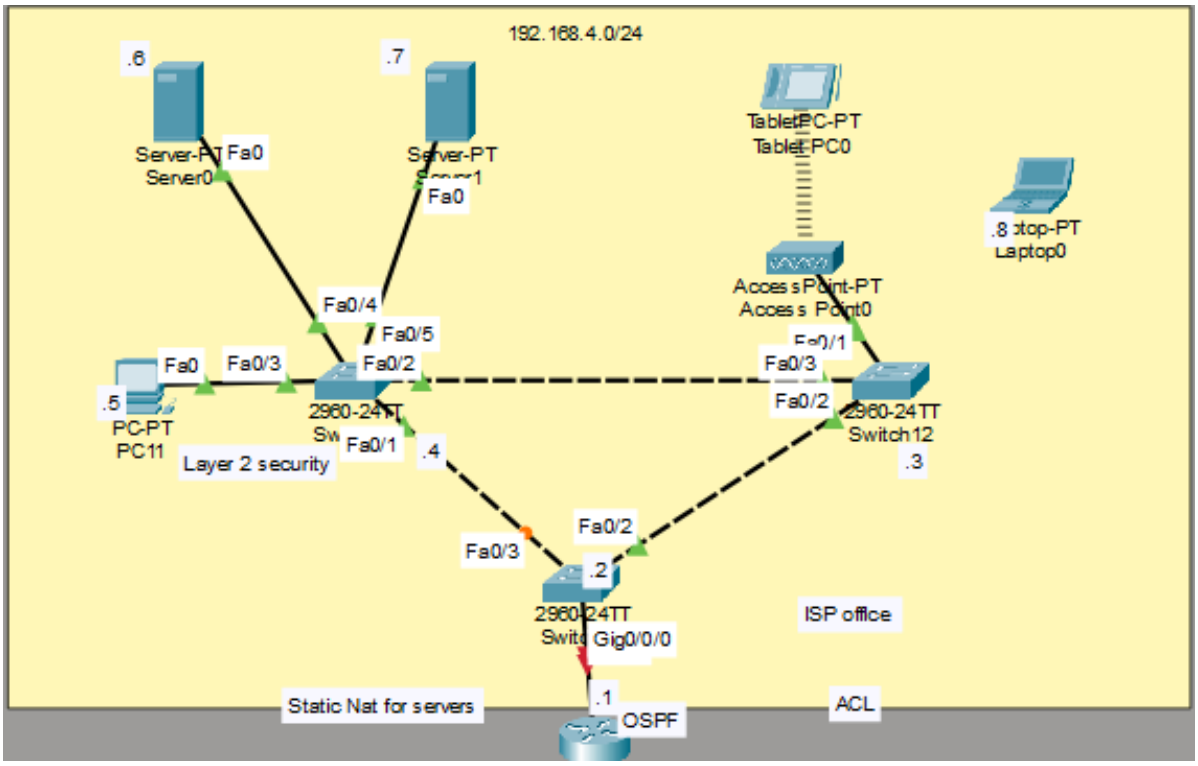
PC0(1)-PC5(1) are split into VLANs 10, 20, and 30 through three regular switches (Switch6, Switch7, and Switch0), which trunk up to Router2. Unlike HQ1, HQ2 has no multilayer switch — Router2 itself routes between the VLANs, and each switch keeps its own management address on VLAN 40.



Users Network (DHCP) – Router1 and Access Switch

Device	Interface	IP Address	Subnet Mask	Wildcard Mask
PC6	Fa0	192.168.3.3	255.255.255.0	0.0.0.255
PC7	Fa0	192.168.3.4	255.255.255.0	0.0.0.255
PC8	Fa0	192.168.3.5	255.255.255.0	0.0.0.255
PC9	Fa0	192.168.3.6	255.255.255.0	0.0.0.255
PC10	Fa0	192.168.3.7	255.255.255.0	0.0.0.255
Access Switch	VLAN 1 (Mgmt)	192.168.3.2	255.255.255.0	0.0.0.255
Router1 (ISR4331)	Gig0/0/0	192.168.3.1	255.255.255.0	0.0.0.255

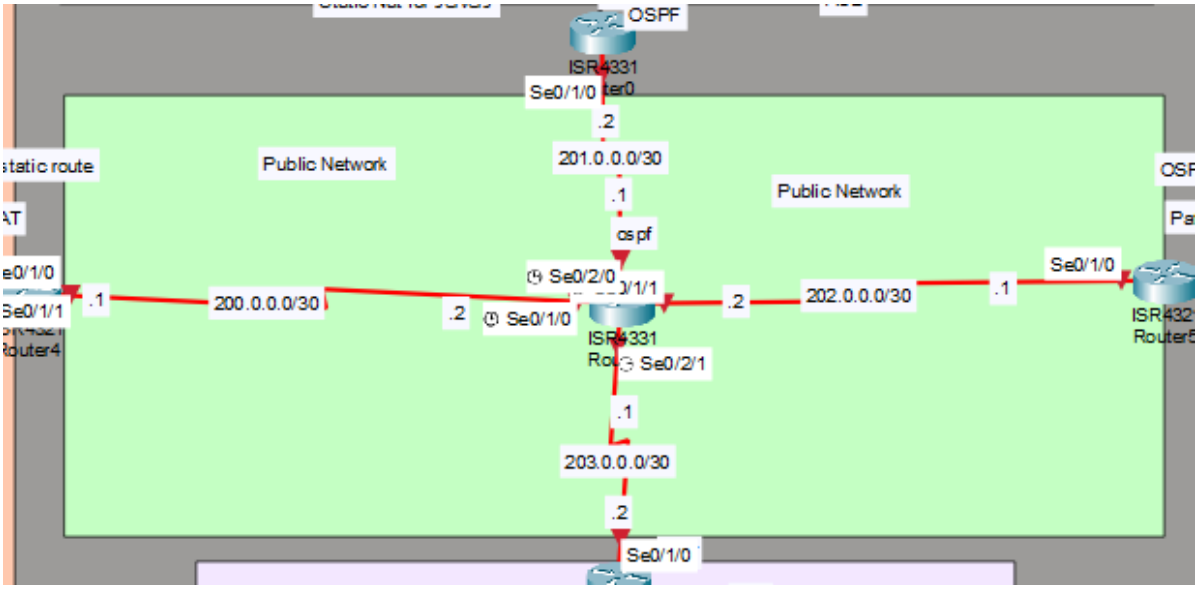
This is the Users Network, 192.168.3.0/24. PC6-PC10 connect through one switch to Router1, which gives them addresses dynamically using DHCP. Router1 also runs OSPF, so this subnet is reachable from HQ1 and HQ2.



ISP Office – Servers, Wireless & Access Switches

Device	Interface	IP Address	Subnet Mask	Wildcard Mask
PC11	Fa0	192.168.4.5	255.255.255.0	0.0.0.255
Server0	Fa0	192.168.4.6	255.255.255.0	0.0.0.255
Server1	Fa0	192.168.4.7	255.255.255.0	0.0.0.255
Laptop0	Wireless	192.168.4.8	255.255.255.0	0.0.0.255
Switch (Layer 2 Security)	VLAN 1 (Mgmt)	192.168.4.2	255.255.255.0	0.0.0.255
Switch12	VLAN 1 (Mgmt)	192.168.4.3	255.255.255.0	0.0.0.255
Uplink Switch	VLAN 1 (Mgmt)	192.168.4.4	255.255.255.0	0.0.0.255
Router (ISP Office)	Gig0/0/0	192.168.4.1	255.255.255.0	0.0.0.255

This is the ISP Office, 192.168.4.0/24. PC11 and the two servers connect to a switch with port security enabled. A second switch connects an Access Point for wireless devices. Both switches trunk to a switch that connects to the router. The router runs OSPF, uses Static NAT for the servers, and has an ACL to control access.



WAN Core – OSPF Backbone (Router0, Router4, Router5 & Router6)

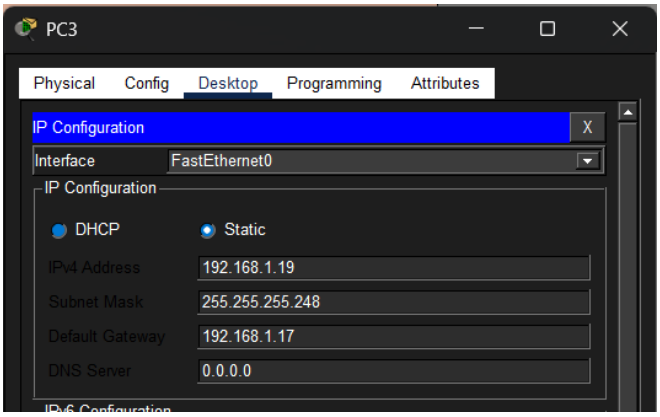
Device	Interface	IP Address	Subnet Mask	Wildcard Mask
Router0 (ISR4331)	Se0/1/0	201.0.0.2	255.255.255.252	0.0.0.3
Router4 (ISR4321)	Se0/1/1	200.0.0.1	255.255.255.252	0.0.0.3
Router6 (ISR4331)	Se0/2/0	201.0.0.1	255.255.255.252	0.0.0.3
Router6 (ISR4331)	Se0/1/0	200.0.0.2	255.255.255.252	0.0.0.3
Router6 (ISR4331)	Se0/1/1	202.0.0.2	255.255.255.252	0.0.0.3
Router6 (ISR4331)	Se0/2/1	203.0.0.1	255.255.255.252	0.0.0.3
Router5 (ISR4321)	Se0/1/0	202.0.0.1	255.255.255.252	0.0.0.3
Router7	Se0/1/0	203.0.0.2	255.255.255.252	0.0.0.3

This is the WAN core that connects HQ1, HQ2, and the outside networks using serial links and OSPF. Router6 is in the center with four serial links: to Router0 (201.0.0.0/30), Router4 (200.0.0.0/30), Router5 (202.0.0.0/30), and Router7 (203.0.0.0/30).

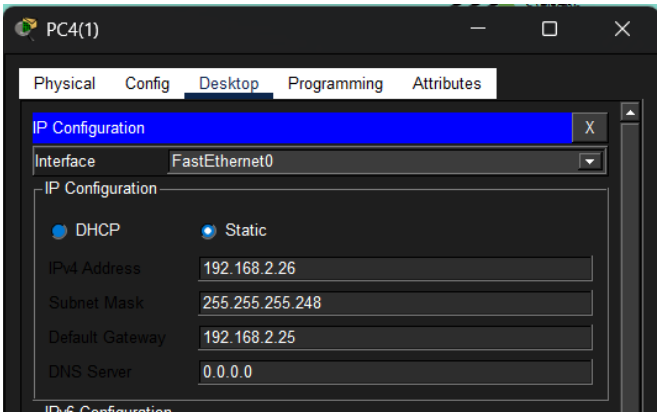
End-Device Addressing

This part shows how each end device got its IP address. HQ1, HQ2, and the ISP Office use static addressing, while the Users Network uses DHCP from Router1.

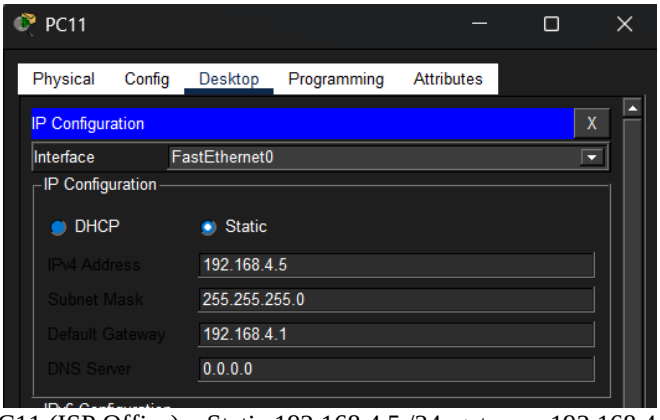
Static Addressing – HQ1, HQ2 & ISP Office



PC3 (HQ1) – Static 192.168.1.19 /29, gateway 192.168.1.17



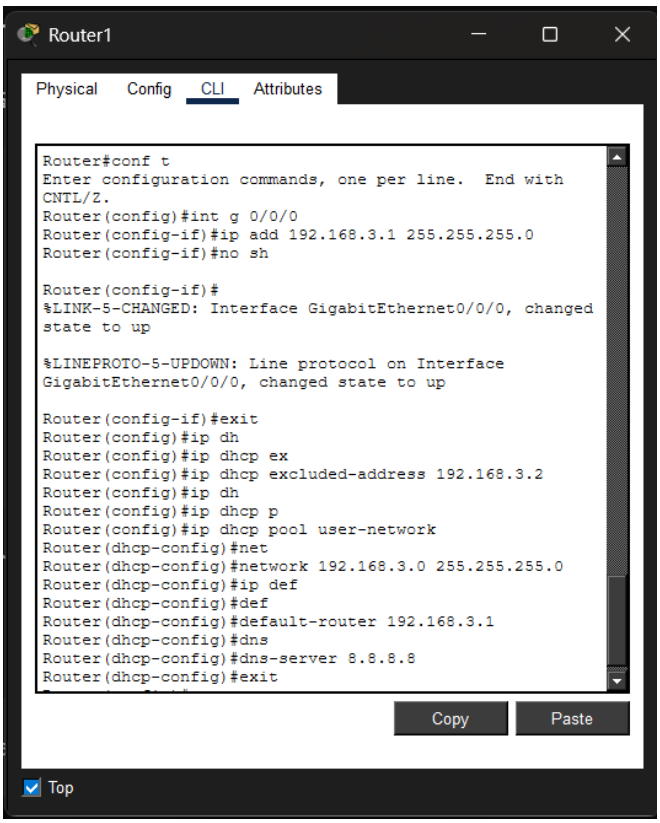
PC4(1) (HQ2) – Static 192.168.2.26 /29, gateway 192.168.2.25



PC11 (ISP Office) – Static 192.168.4.5 /24, gateway 192.168.4.1

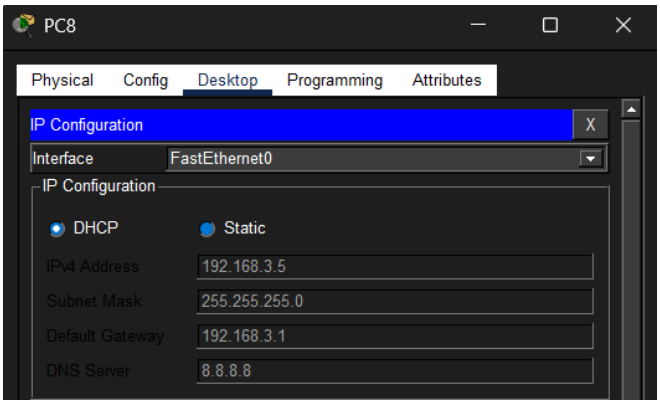
Each PC has Static IP Configuration, with the address, mask, and gateway typed in by hand to match the table. PC3 is in HQ1's VLAN 10, PC4(1) is in HQ2's VLAN 20, and PC11 is in the ISP Office network.

DHCP Addressing – Users Network



Router1 (config) – DHCP pool "user-network" for 192.168.3.0/24

On Router1, Gig0/0/0 got the gateway address 192.168.3.1/24, then a DHCP pool named user-network was created for 192.168.3.0/24. It excludes 192.168.3.2 for the switch's management address, and gives every PC the gateway 192.168.3.1 and DNS 8.8.8.8.



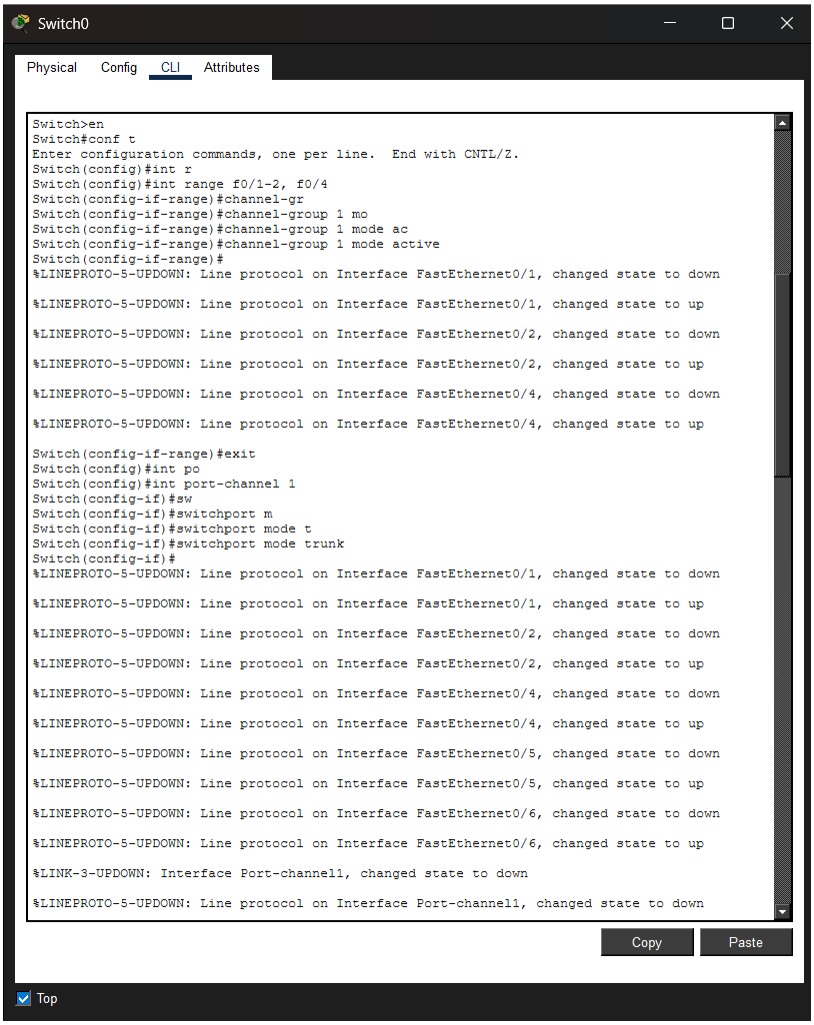
PC8 (Users Network) – DHCP assigned 192.168.3.5, gateway 192.168.3.1, DNS 8.8.8.8

PC8's IP Configuration confirms DHCP is working: DHCP is selected, and the fields are filled in automatically with 192.168.3.5, mask 255.255.255.0, gateway 192.168.3.1, and DNS 8.8.8.8.

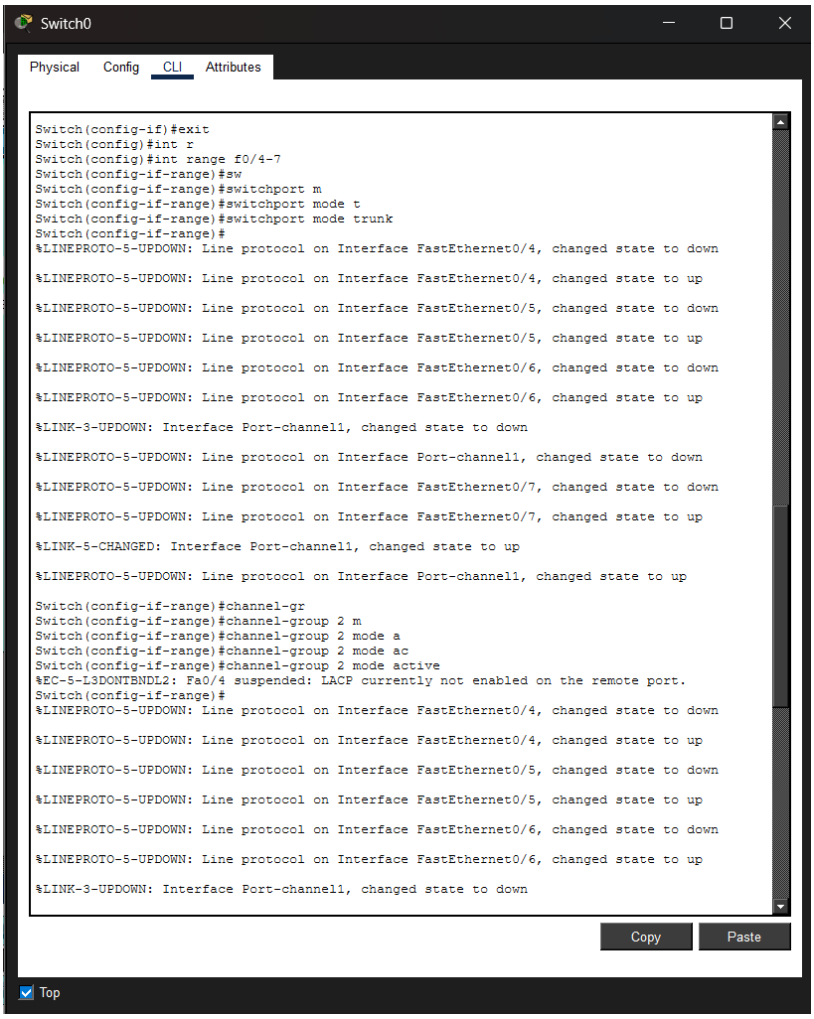
VLANs & EtherChannel (LACP)

This part covers VLANs and EtherChannel. VLANs split the access ports into separate broadcast domains, and EtherChannel (LACP) bundles the links between switches into one connection, ensuring network resiliency if a single link fails.

EtherChannel (LACP)

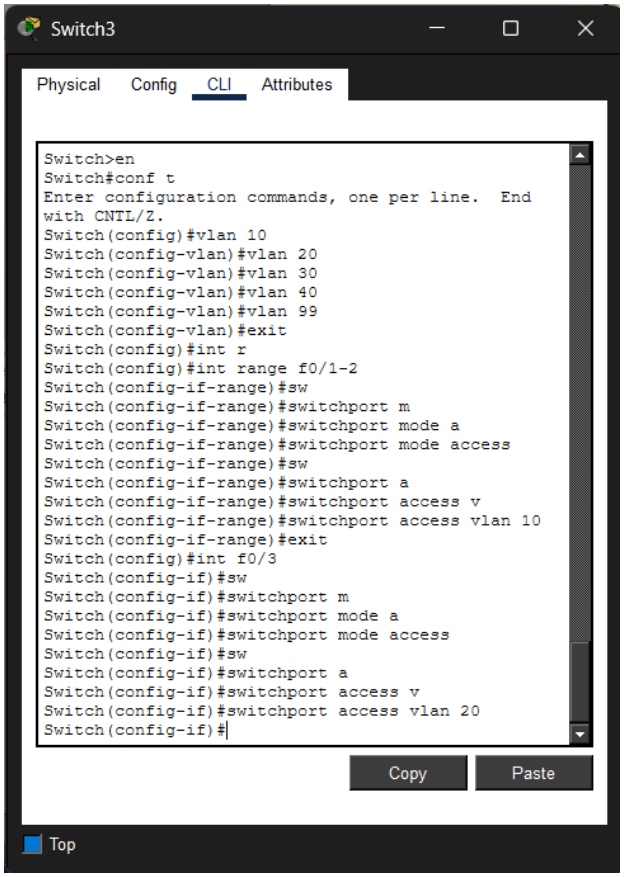


Switch0 — Fa0/1, Fa0/2 and Fa0/4 joined into channel-group 1 (LACP active); Port-channel1 then set to switchport mode trunk

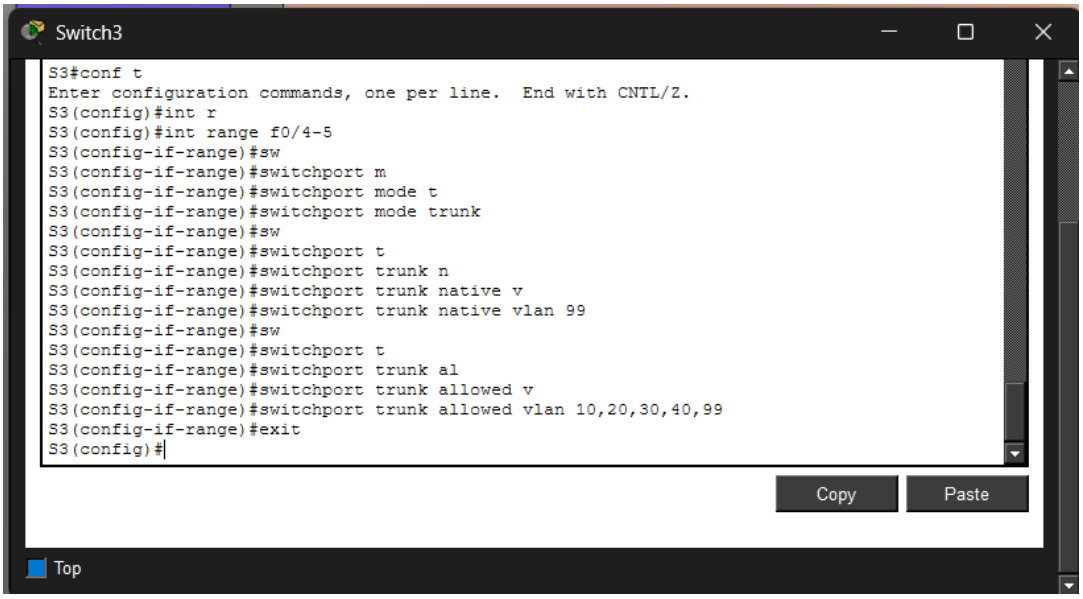


Switch0 — Fa0/4-7 set to trunk, then a second bundle (channel-group 2, LACP active) started on the same range; Fa0/4 shows suspended while waiting for the remote side

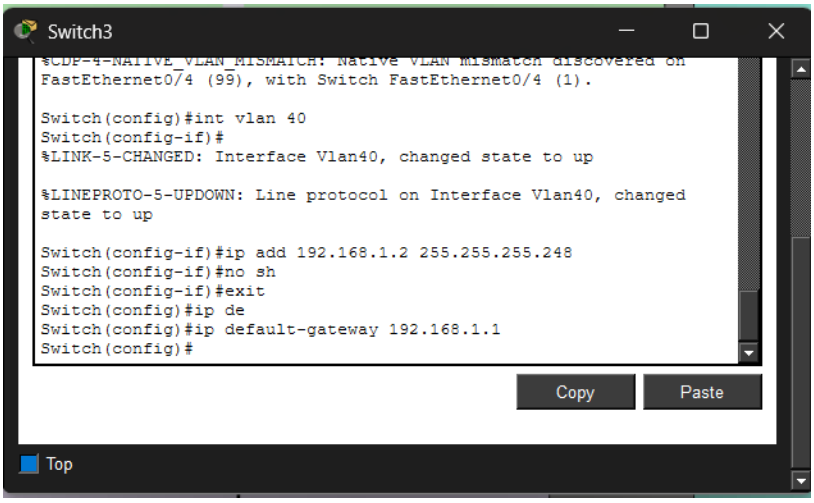
VLANs – HQ1 & HQ2 Access Switches



Switch3 (HQ1) – VLANs 10, 20, 30, 40 & 99 created; Fa0/1-2 set to access VLAN 10, Fa0/3 set to access VLAN 20



Switch3 (HQ1) – Fa0/4-5 set to trunk with native VLAN 99 and allowed VLANs 10,20,30,40,99



Switch3 (HQ1) – Management address 192.168.1.2/29 on VLAN 40, gateway 192.168.1.1

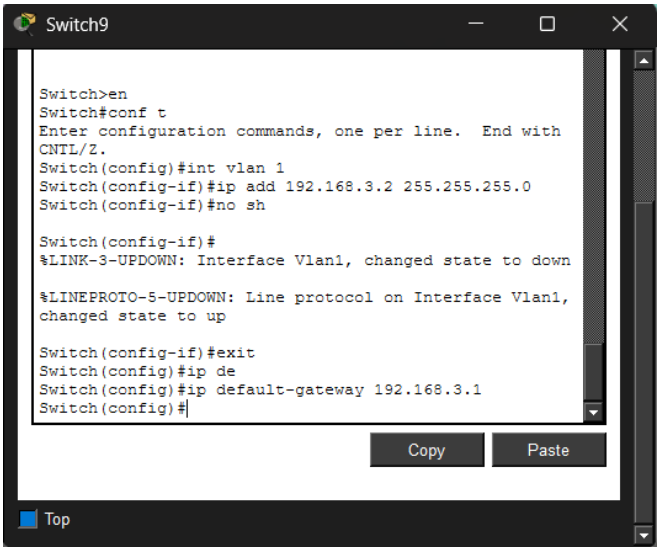
On Switch3, VLANs 10, 20, 30, 40, and 99 were created, then Fa0/1-2 were set to access VLAN 10 and Fa0/3 to VLAN 20. Fa0/4-5 were set to trunk mode with native VLAN 99, allowing all VLANs.VLAN 40 is used for management: Switch3 gets 192.168.1.2/29 with gateway 192.168.1.1.



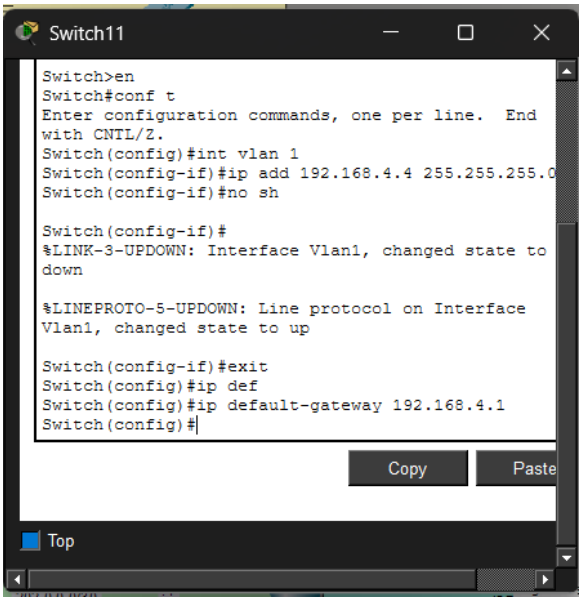
Switch0 (HQ2) – Same VLAN, trunk, and management pattern: VLANs 10/20/30/40/99, Fa0/3 trunk (native VLAN 99, allowed 10,20,30,40,99), management address 192.168.2.4/29 on VLAN 40, gateway 192.168.2.1

The HQ2 switch (Switch0) follows the same pattern: VLANs 10, 20, 30, 40, and 99 are created, Fa0/3 is trunked with native VLAN 99, and its management address 192.168.2.4/29 sits on VLAN 40 with gateway 192.168.2.1.

Management VLAN – Users Network & ISP Office



Switch9 (Users Network access switch) – Management address 192.168.3.2/24 on VLAN 1, gateway 192.168.3.1



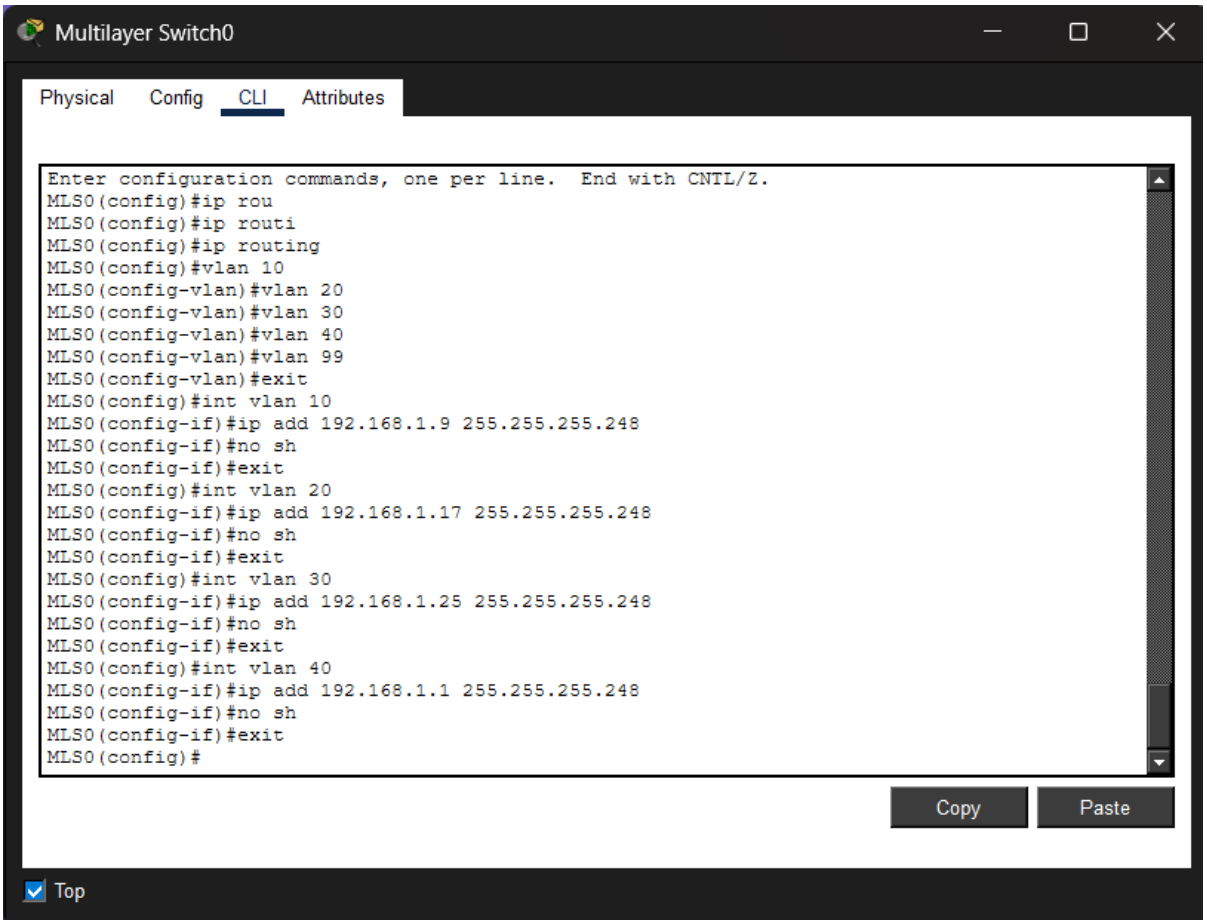
Switch11 (ISP Office uplink switch) – Management address 192.168.4.4/24 on VLAN 1, gateway 192.168.4.1

The Users Network and ISP Office don't need multiple VLANs, so both switches keep VLAN 1 with a management address: Switch9 gets 192.168.3.2/24, gateway 192.168.3.1, and Switch11 gets 192.168.4.4/24, gateway 192.168.4.1.

MLS Routing & Gateways

This part turns multilayer switch into a router: ip routing is enabled, each VLAN gets an SVI as its gateway, and one port becomes a routed link to the site's router.

Multilayer Switch0 (HQ1)



Multilayer Switch0 – ip routing enabled; VLANs 10,20,30,40,99 created

```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int g0/1
Switch(config-if)#no sw
Switch(config-if)#no switchport
Switch(config-if)#ip add 192.168.1.33 255.255.255.252
Switch(config-if)#no sh
```

Multilayer Switch0 – Gig0/1 converted to a routed port (no switchport) with IP 192.168.1.33/30 toward Router3

First, ip routing is enabled so the switch can route between VLANs. The same VLANs (10, 20, 30, 40, 99) are created,: VLAN 10 → 192.168.1.9, VLAN 20 → 192.168.1.17, VLAN 30 → 192.168.1.25, VLAN 40 → 192.168.1.1. Finally, Gig0/1 is set as a routed port with IP 192.168.1.33/30, connecting to Router3.

Router2 (HQ2)



Router2 (HQ2) – Gig0/0/0 enabled; subinterfaces g0/0/0.10, .20, .30 and .40 created with dot1Q encapsulation, carrying gateway addresses 192.168.2.9, 192.168.2.17, 192.168.2.25 and 192.168.2.1

HQ2 is different: instead of the multilayer switch, Router2 acts as a router-on-a-stick. Gig0/0/0 is split into four, one per VLAN, each with 802.1Q encapsulation and a gateway address: g0/0/0.10 → 192.168.2.9, g0/0/0.20 → 192.168.2.17, g0/0/0.30 → 192.168.2.25, g0/0/0.40 → 192.168.2.1. One trunk link carries all four VLANs to Router2, which routes between them.

OSPF Routing & Default Route

This part enables OSPF on every router and multilayer switch so all sites learn each other's routes automatically.

Multilayer Switch0 (HQ1) — OSPF

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int g0/1
Switch(config-if)#no sw
Switch(config-if)#no switchport
Switch(config-if)#ip add 192.168.1.33 255.255.255.252
Switch(config-if)#no sh
Switch(config-if)#exit
Switch(config)#router ospf
% Incomplete command.
Switch(config)#router ospf 1
Switch(config-router)#net
Switch(config-router)#network 192.168.1.0 0.0.0.7 area 0
Switch(config-router)#net
Switch(config-router)#network 192.168.1.8 0.0.0.7 area 0
Switch(config-router)#net
Switch(config-router)#network 192.168.1.16 0.0.0.7 area 0
Switch(config-router)#net
Switch(config-router)#network 192.168.1.24 0.0.0.7 area 0
Switch(config-router)#net
Switch(config-router)#network 192.168.1.32 0.0.0.3 area 0
Switch(config-router)#pass
Switch(config-router)#passive-interface v
Switch(config-router)#passive-interface vlan 10
Switch(config-router)#pas
Switch(config-router)#passive-interface v
Switch(config-router)#passive-interface vlan 20
Switch(config-router)#pas
Switch(config-router)#passive-interface in
Switch(config-router)#passive-interface v
Switch(config-router)#passive-interface vlan 30
Switch(config-router)#pas
Switch(config-router)#passive-interface v
Switch(config-router)#passive-interface vlan 40
Switch(config-router)#exit
```

Multilayer Switch0 — router ospf 1; networks for VLANs 10/20/30 and the Router3 link advertised; VLANs 10, 20, 30, 40 set passive-interface

Multilayer Switch0 advertises the four HQ1 subnets and the link to Router3, with all VLAN interfaces set passive.

Router3 (HQ1) — OSPF

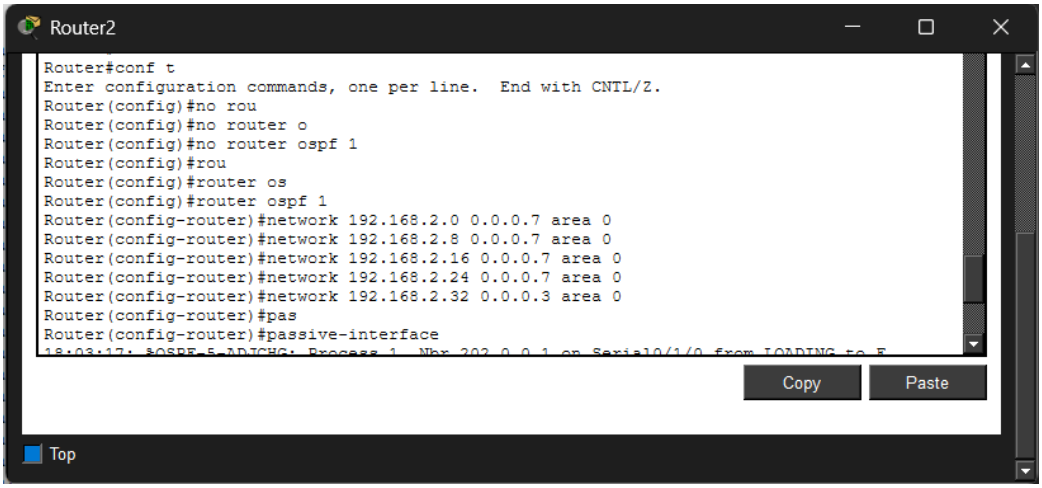
```
Router(config)#router ospf 1
Router(config-router)#net
Router(config-router)#network 192.168.1.32 0.0.0.3 area 0
Router(config-router)#net
Router(config-router)#network 192.168.1.36 0.0.0.3 area 0
21:07:00: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.33 on
GigabitEthernet0/0/0 from LOADING to FULL, Loading Done

Router(config-router)#exit
Router(config)#
```

Router3 — networks 192.168.1.32/30 (to MLS0) and 192.168.1.36/30 (to the WAN link) added; adjacency with Multilayer Switch0 (192.168.1.33) reaches FULL

Router3 adds both HQ1 links into OSPF.

Router2 (HQ2) — OSPF



Router2 — OSPF process re-created (router ospf 1); networks 192.168.2.0/29, 192.168.2.8/29, 192.168.2.16/29, 192.168.2.24/29 and 192.168.2.32/30 advertised in area 0; adjacency with Router5 (202.0.0.1) on Se0/1/0 reaches FULL

Router2 advertises all four HQ2 subnets and the serial uplink (192.168.2.32/30) into OSPF area 0, and forms a FULL adjacency with Router5 over Se0/1/0.

Router5 — HQ2 Uplink & WAN Core

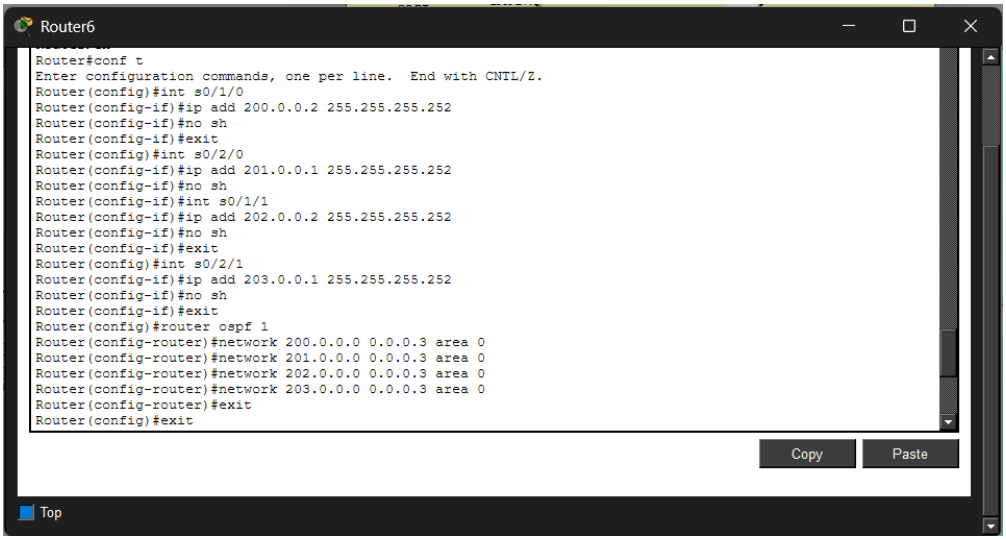
```
R5(config)#rou
R5(config)#router osp
R5(config)#router ospf 1
R5(config)#router ospf 1
R5(config-router)#passive-interface serial 0/1/0
R5(config-router)#net
R5(config-router)#network 192.168.2.32 0.0.0.3 area 0
R5(config-router)#
18:12:13: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.33 on Serial0/1/1 from LOADING to FULL,
Loading Done

R5(config-router)#network 202.0.0.0 0.0.0.3 area 0
```

Router5 — router ospf 1; Se0/1/0 (toward Router6) set passive-interface; networks 192.168.2.32/30 and 202.0.0.0/30 in area 0; adjacency with Router2 (192.168.2.33) on Se0/1/1 reaches FULL

Router5 connects HQ2 to the WAN core and forms a FULL adjacency with Router2. Its Se0/1/0 interface toward Router6 is set passive, so OSPF hellos are only sent on the HQ2-facing link.

Router6 — WAN Core



Router6 — Se0/1/0, Se0/2/0, Se0/1/1 and Se0/2/1 addressed; router ospf 1 with networks 200.0.0.0/30, 201.0.0.0/30, 202.0.0.0/30 and 203.0.0.0/30, all area 0

Router6 adds all four serial links into OSPF area 0, making it the hub that connects Router0, Router4, Router5, and Router1.

Router4 — Outside Edge & Default Route

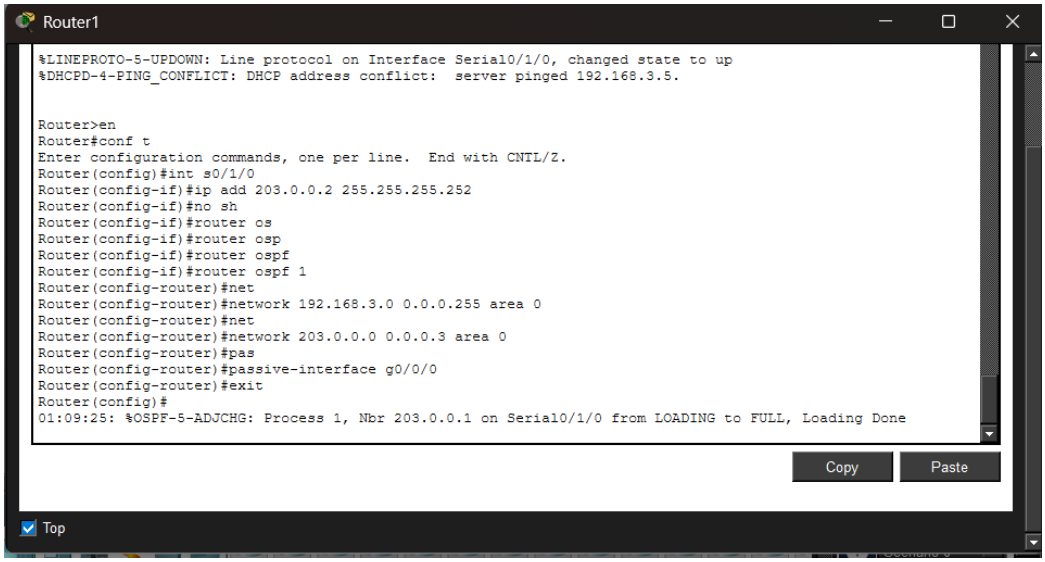
```
R4(config)#ip route 0.0.0.0 0.0.0.0 s0/1/1
%Default route without gateway, if not a point-to-point interface, may impact performance
R4(config)#rou
R4(config)#router os
R4(config)#router ospf 1
R4(config-router)#net
R4(config-router)#network 192.168.1.36 0.0.0.3 area 0
R4(config-router)#n
17:43:06: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.37 on Serial0/1/0 from LOADING to FULL,
Loading Done
et
R4(config-router)#network 200.0.0.0 0.0.0.3 area 0
R4(config-router)#def
R4(config-router)#default-information or
R4(config-router)#default-information originate
R4(config-router)#
17:43:36: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.0.1 on Serial0/1/1 from LOADING to FULL,
Loading Done

R4(config-router)#pas
R4(config-router)#passive-interface s0/1/1
```

Router4 — static default route (ip route 0.0.0.0 0.0.0.0 s0/1/1) pointed out Se0/1/1 toward the outside network; router ospf 1 with networks 192.168.1.36/30 and 200.0.0.0/30 in area 0; adjacency with Router3 (192.168.1.37) reaches FULL; default-information originate and passive-interface s0/1/1

Router4 is the edge toward the outside network: it holds the static default route and also runs OSPF, so the route can be shared with the rest of the network using default-information originate. Its Se0/1/1 interface toward Router6 is set passive, so OSPF hellos are only sent on the HQ1-facing link.

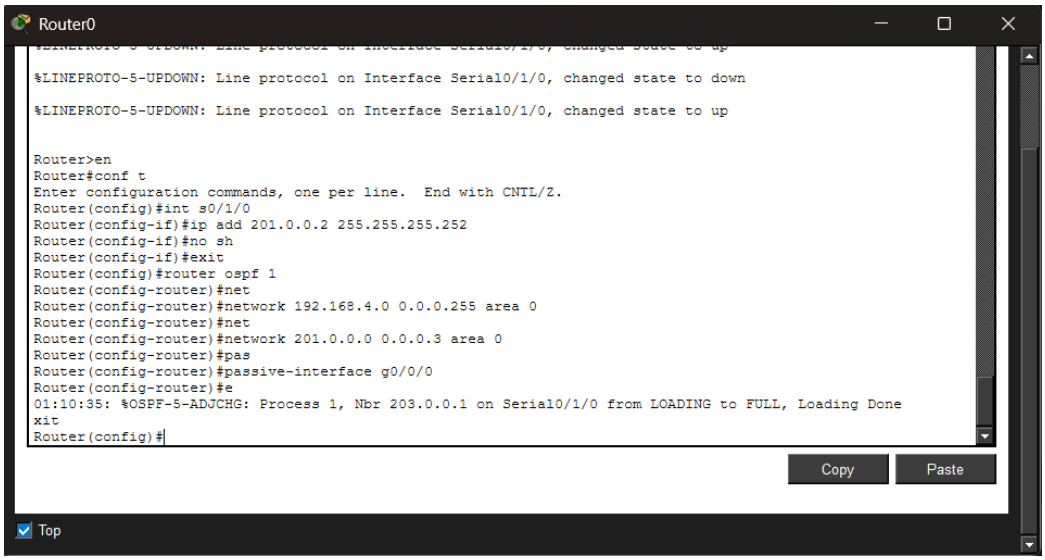
Router1 — Users Network



Router1 — Se0/1/0 (203.0.0.2/30); router ospf 1 with networks 192.168.3.0/24 and 203.0.0.0/30; g0/0/0 (the DHCP LAN interface) set passive-interface

Router1 adds the DHCP-served Users Network into OSPF and keeps its LAN interface passive, so only the WAN link sends OSPF hellos.

Router0 — ISP Office



Router0 — Se0/1/0 (201.0.0.2/30); router ospf 1 with networks 192.168.4.0/24 and 201.0.0.0/30; g0/0/0 set passive-interface

Router0 follows the same pattern: its LAN stays passive while Se0/1/0 carries OSPF into the WAN core.

Dynamic PAT (NAT Overload)

Each router sets its LAN interface as NAT inside and its WAN interface as NAT outside, then overloads its local subnet behind one public IP.

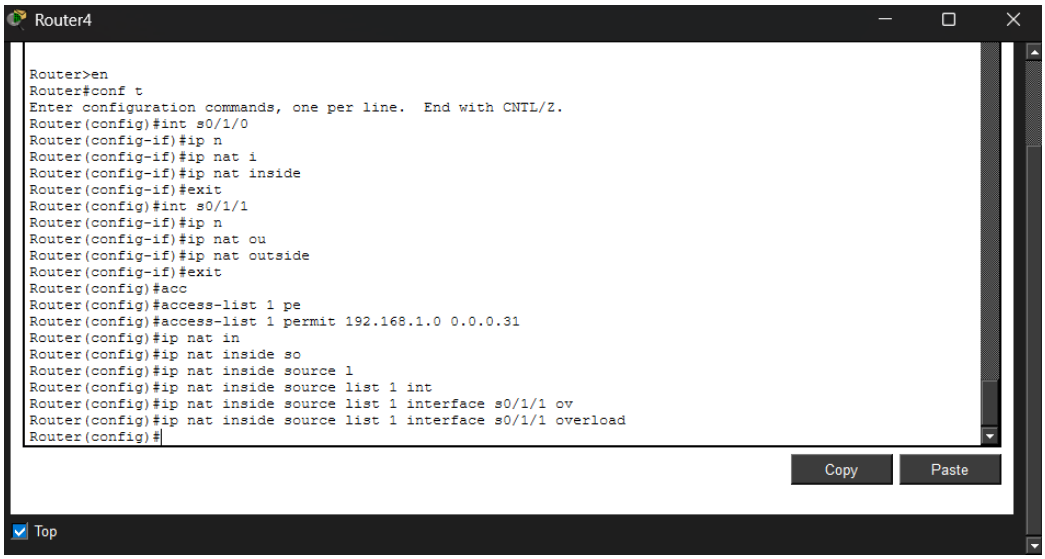
Router1 — Users Network



Router1 — g0/0/0 set NAT inside, Se0/1/0 set NAT outside; access-list 1 permits 192.168.3.0/24; ip nat inside source list 1 interface s0/1/0 overload

Router1 overloads the whole Users Network subnet out through its serial link.

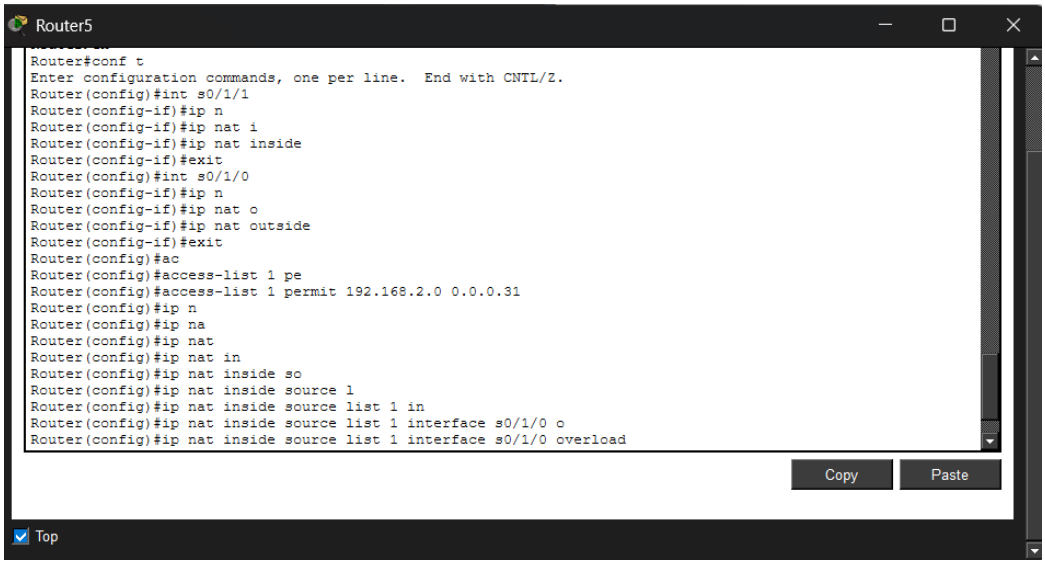
Router4 — HQ1 Outbound



Router4 — Se0/1/0 set NAT inside, Se0/1/1 set NAT outside; access-list 1 permits 192.168.1.0/27; ip nat inside source list 1 interface s0/1/1 overload

Router4 overloads HQ1's subnet out toward the outside network.

Router5 — HQ2 Outbound

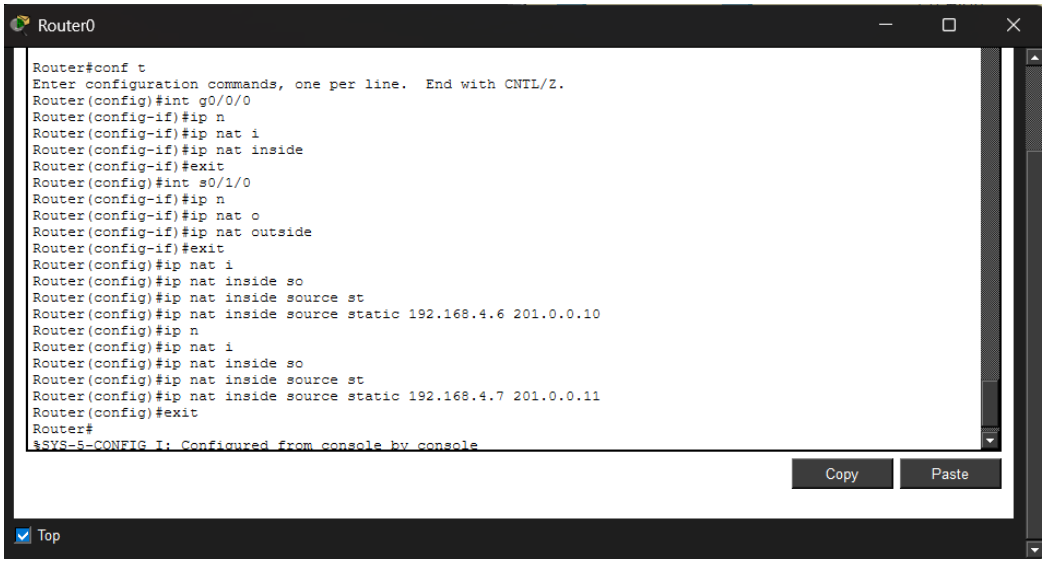


Router5 — Se0/1/1 set NAT inside, Se0/1/0 set NAT outside; access-list 1 permits 192.168.2.0/27; ip nat inside source list 1 interface s0/1/0 overload

Router5 does the same for HQ2, overloading its subnet through the link toward Router6.

Static NAT for Servers

Router0 — ISP Office Servers



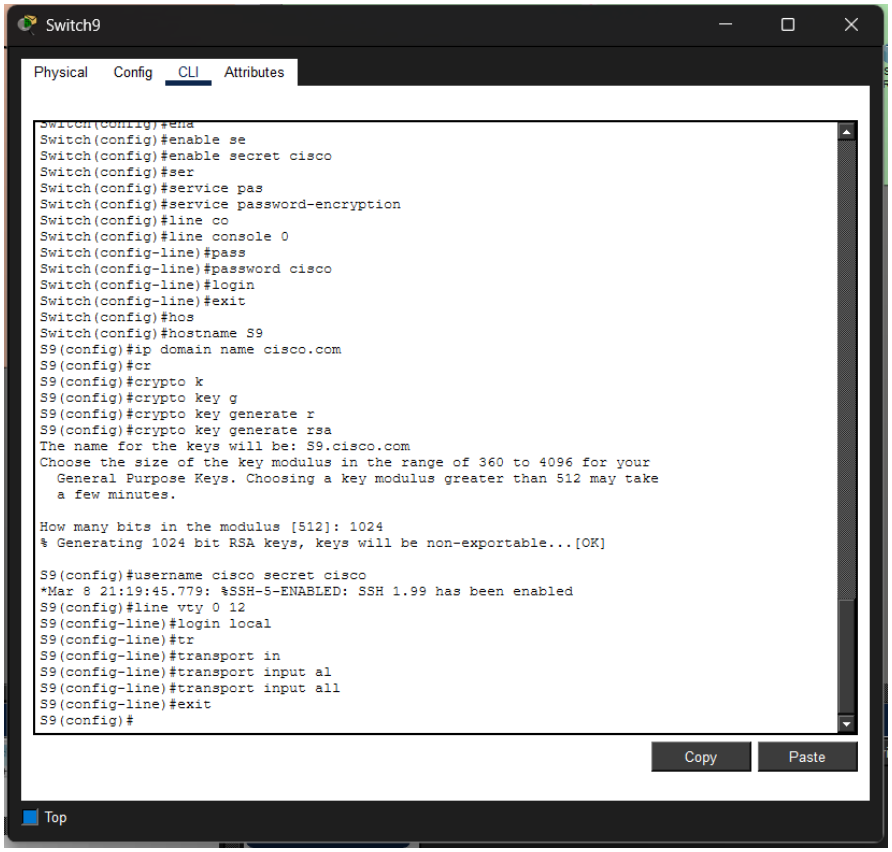
Router0 — g0/0/0 set NAT inside, Se0/1/0 set NAT outside; static mappings 192.168.4.6 → 201.0.0.10 and 192.168.4.7 → 201.0.0.11

Router0 gives the two servers fixed public addresses so they're always reachable from outside on the same IP.

Security & Remote Control

This part locks down device access: passwords are encrypted, and each device gets its own hostname and domain name.

Switch9 — SSH & Passwords



Switch9 — enable secret and console password set, service password-encryption on; hostname S9, domain cisco.com, 1024-bit RSA keys generated; user cisco added; vty lines set to login local with SSH (transport input all)

Switch9 gets an encrypted enable password, a console password, and RSA keys so it can run SSH. The vty lines use a local username instead of a single shared password.

Router1 — SSH & Passwords



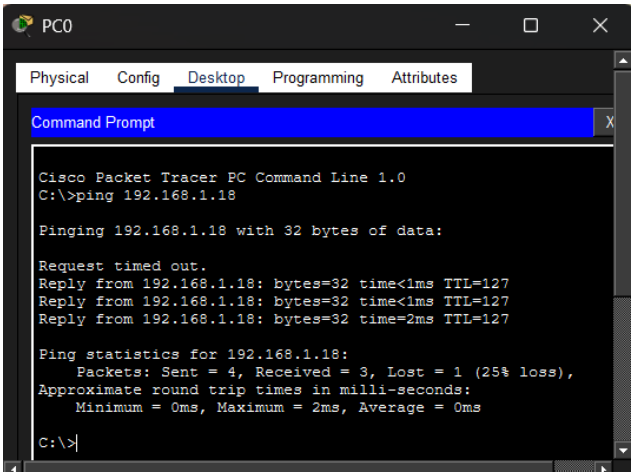
Router1 — same setup as Switch9 (hostname R1, domain cisco.com, RSA keys, user cisco, vty login local with SSH), then the config is saved with copy running-config startup-config

Router1 is configured the same way as Switch9, and the running configuration is saved to startup-config so the settings survive a reboot.

Verification

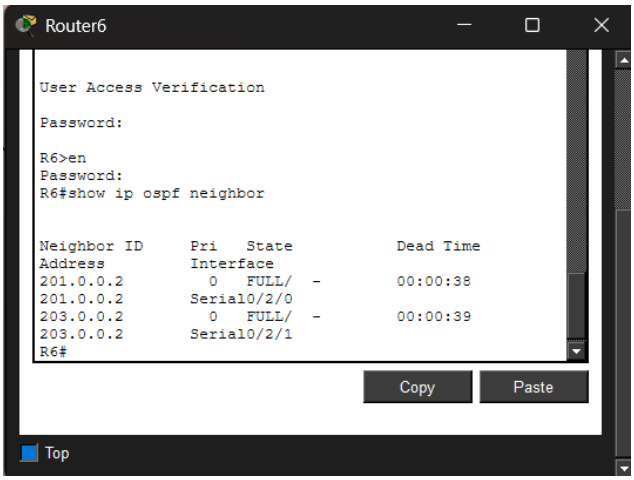
This part checks that the network works as designed. Each screenshot below tests one feature: inter-VLAN routing, OSPF adjacencies, Static NAT reachability, Dynamic PAT translations, and SSH remote access.

Inter-VLAN Routing — PC0 to PC2 (HQ1)



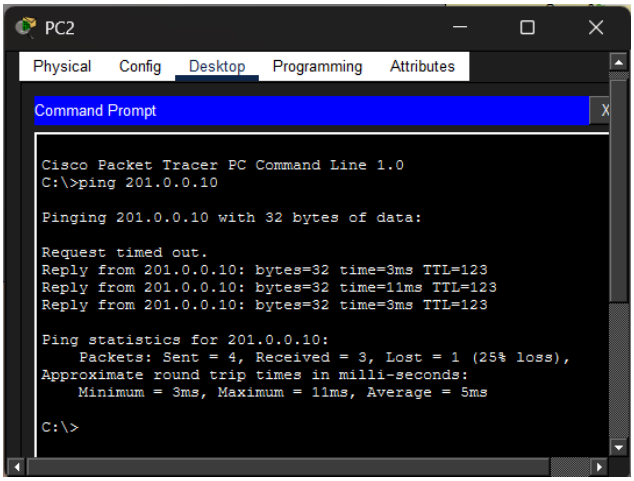
PC0 (VLAN 10, 192.168.1.10) pings PC2 (VLAN 20, 192.168.1.18) through Multilayer Switch0 (TTL=127). The first request times out while ARP resolves, then the replies succeed (3 of 4).

OSPF Adjacencies — Router6 (WAN Core)



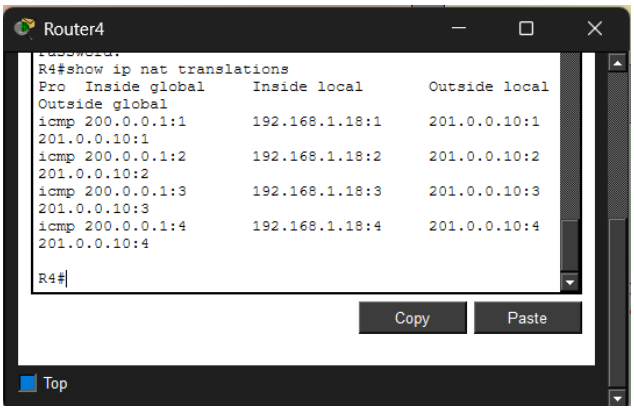
Router6 — show ip ospf neighbor: FULL adjacencies with 201.0.0.2 (Router0, Se0/2/0) and 203.0.0.2 (Router1, Se0/2/1).

Static NAT Reachability — PC2 to Server0



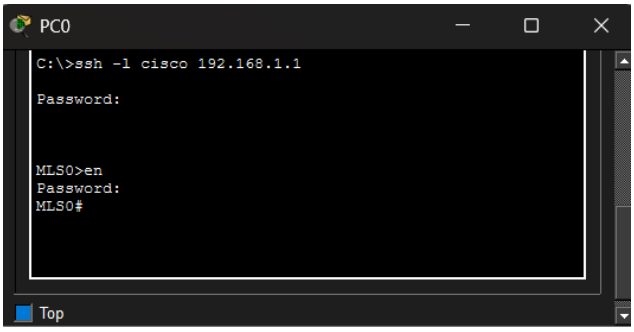
PC2 (HQ1) pings 201.0.0.10, the public address mapped to Server0 by Static NAT. The first request times out, as on a first ping, then 3 of 4 replies succeed.

Dynamic PAT — Router4 Translations



Router4 — show ip nat translations: PC2's ICMP traffic (inside local 192.168.1.18) is translated to 200.0.0.1 (inside global) on its way to 201.0.0.10, confirming Dynamic PAT.

Remote Access — SSH from PC0 to MLS0



PC0 runs ssh -l cisco 192.168.1.1 (the VLAN 40 address of Multilayer Switch0), logs in, and reaches the MLS0# privileged prompt.